

BodyQuest – Detailed Level Design

1. Game Overview

This document defines a full level design for the game **BodyQuest**.

The goal is to describe four concrete levels – Tutorial, Easy, Medium, and Hard – with enough detail so that a programmer can implement the levels directly.

Each level specification includes: narrative setup, gameplay objectives, detailed content (organs visited, mini-games, enemies, stats), win/lose conditions, difficulty rationale, and a textual diagram of the internal body map.

2. Level 1 – Tutorial (Beginner)

1. Narrative Setup

- The player is a tiny explorer entering the human body for the first time.
- The body belongs to a healthy teenager on a normal day.
- Mood: playful and educational – no real danger yet.

2. Player Objectives

- Learn the basic stats: Energy, Mood, Health Bar.
- Complete simple tasks in two organs: Stomach and Heart.
- Understand how one food choice affects Energy and Mood.

3. Detailed Content

3.1 Organs and Areas

- Mouth: entry point, no gameplay yet – only a short scripted intro.
- Esophagus corridor: straight corridor connecting Mouth to Stomach.
- Stomach chamber: first interactive area, shows food processing.
- Heart chamber: second area, shows blood pumping animation.

3.2 Main Mechanics Introduced

- Choosing one meal from three options at the start:
 - 1) Balanced meal: rice + chicken + salad – medium-fast Energy increase, no Mood penalty.
 - 2) Fast food: big burger + fries – large immediate Energy spike, small future drop.

3) Candy-only: chocolate and soda – big short Energy spike, strong later crash and Mood penalty.

- Simple digestion mini-game in the Stomach:

- Player must drag food particles into correct “enzyme zones” (proteins, carbs, fats).

- Failure just slows Energy gain; cannot lose the level here.

- Heart encounter:

- Player moves through a short tunnel, dodging 2–3 slow-moving red blood cells as obstacles.

- Reaching the end fills the Health Bar slightly.

3.3 Example Gameplay Flow

1) Player selects meal.

2) Short cutscene of food going down esophagus.

3) In Stomach, digestion mini-game appears for 30–45 seconds.

4) After success, Energy bar animates up based on meal choice.

5) Player walks through a short vein corridor to Heart.

6) In Heart, the player must time their movement with a pumping animation to cross a moving platform.

4. Win / Lose Conditions

Win:

- Successfully complete the digestion mini-game (even with mistakes).

- Reach the Heart exit area.

Lose:

- No hard fail in tutorial – if the player “fails” the mini-game heavily, they simply receive less Energy and see a warning tooltip explaining why.

5. Difficulty Rationale

- No real danger or strict timers – the goal is to teach core systems.

- Choices are meaningful but consequences are gentle and mostly visual.

- Only two organs, small number of moving obstacles.

6. Level Diagram (Textual)

BODY MAP – LEVEL 1

[Outside World]

|

v

[Mouth]

|

v

[Esophagus]

|

v

[Stomach] --(Digestion mini-game)---> [Vein Corridor] ---> [Heart] ---> [Exit]

3. Level 2 – Easy

1. Narrative Setup

- Same body, later that day – the teenager is now studying for a test.
- Mood: slightly stressed, requires focus and physical posture.

2. Player Objectives

- Visit Brain and Muscle zones.
- Maintain enough Energy and Mood to complete a focus puzzle and a rhythm exercise.
- Experience the effect of combining food and small exercise.

3. Detailed Content

3.1 Organs and Areas

- Brain – “Control Room” with many glowing connections.
- Muscle Zone – represented as a field of fibers that expand and contract.

3.2 Systems and Mini-Games

Brain – Focus Puzzle:

- The player is shown a small pattern of 6 glowing nodes for 3 seconds.
- Then the nodes turn off, and the player must re-activate them in the same order.
- 3 attempts allowed.
- Difficulty parameters:
 - Number of nodes: 6.
 - Maximum pattern length: 4 steps.
 - Time to input: 20 seconds.

Muscle – Rhythm Exercise:

- The player runs along a flat “muscle fiber” track.
- Beat markers appear on the ground; the player presses a key when standing on them.
- Missing too many beats reduces the effectiveness of the exercise.

3.3 Events and Choices

- Before going to Brain, the player gets to pick a small snack:
 - 1) Fruit snack – small Energy boost, Mood boost.
 - 2) More candy – temporary Energy spike, Mood drop afterwards.
- After finishing Brain, the player is slightly mentally tired; the game suggests going to Muscle zone for a short “stretch”.

3.4 Example Gameplay Flow

- 1) From Heart, a new path to Brain opens.
- 2) Player enters Brain, sees pattern puzzle mini-game.
- 3) Success grants:
 - +Focus state (temporary Mood buff).
 - Slight Energy cost (mental strain).
- 4) Player is advised to head to Muscle Zone to “shake off” the tension.
- 5) In Muscle mini-game, hitting enough beats gives:
 - +Mood boost, +Health Bar.
- 6) If the player played badly:
 - Gains less benefit, and receives hint text about the value of balanced movement.

4. Win / Lose Conditions

Win:

- Completing at least one Brain puzzle successfully.
- Reaching the end of Muscle track.

Partial success:

- If the Brain puzzle is failed all 3 times:
 - Player continues, but receives lower Mood buff and an educational tooltip.

Lose:

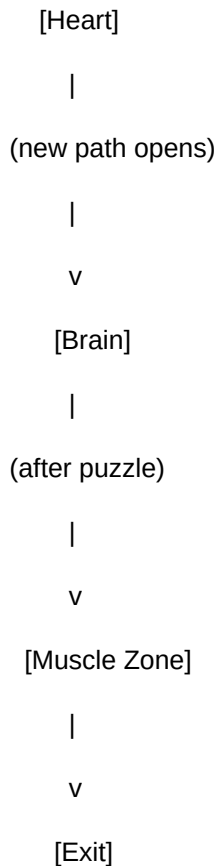
- Can only fail by letting Energy drop to zero by choosing very bad snack options and then taking too long – unlikely but possible.

5. Difficulty Rationale

- Two separate mini-games requiring memory and rhythm – more demanding than Level 1.
- Still low risk; main challenge is cognitive, not survival.
- Encourages understanding that mental and physical activities affect each other.

6. Level Diagram (Textual)

BODY MAP – LEVEL 2



4. Level 3 – Medium

1. Narrative Setup

- Next day: the teenager has eaten somewhat unbalanced meals and slept poorly.
- The body is under mild stress; an infection event starts in the lower abdomen.

2. Player Objectives

- Deal with a small infection using the Immune System.
- Help the Liver process toxins from junk food.
- Keep Energy and Health from dropping too low by managing choices and timing.

3. Detailed Content

3.1 Organs and Areas

- Liver – represented as a factory with conveyor belts and “toxin containers”.
- Immune Battlefield – an enlarged area of tissue with invading bacteria.

3.2 Mini-Games and Enemies

Liver – Toxin Cleanup:

- Food history from previous levels determines how many toxins appear.
- Player sees blocks labelled with symbols (e.g., F for Fat, S for Sugar, A for Alcohol-like toxins).
- Conveyor belts move these blocks slowly towards the “Bloodstream” exit.
- The player must drag each block into its matching detox machine before it exits.
- Parameters:
 - 12–16 toxin blocks in total.
 - 3 different machine types.
 - Time per block: roughly 5 seconds.

Immune System – Bacteria Waves:

- The player controls a small “immune cell” avatar.
- 3 short waves of bacteria appear:
 - Wave 1: 5 slow bacteria.
 - Wave 2: 8 medium-speed bacteria.
 - Wave 3: 10 mixed-speed bacteria.
- The player must tag each bacterium by touching it; marked bacteria are then destroyed by auto-attack.

3.3 Resource Effects

- If too many toxins escape the Liver:
 - Health Bar drops significantly.
 - Infection waves in the Immune area become harder (bacteria get small speed boost).
- If the player performs very well in Liver:
 - Health Bar stabilizes, Immune waves are easier.

3.4 Example Gameplay Flow

- 1) Brief cutscene: red warning icons on the Liver and in the lower abdomen.

2) Player goes to Liver first:

- Plays detox mini-game for about 60–90 seconds.

3) Based on performance, parameters for Immune waves are set.

4) Player enters Immune area and fights 3 waves of bacteria.

5) After success, Health Bar increases and a calm music cue plays.

4. Win / Lose Conditions

Win:

- Survive all 3 bacteria waves.
- Prevent Health Bar from reaching zero.

Partial success:

- If many toxins escaped Liver but player still wins Immune battle:
 - End state: Health Bar remains low-medium; educational message about long-term effects.

Lose:

- Health Bar reaching zero at any point during this level.
- Failing to eliminate enough bacteria in wave 3 before a timer ends.

5. Difficulty Rationale

- Two tightly linked systems (Liver and Immune) – mistakes in one level affect the difficulty of the other.
- More enemies (bacteria) and stricter timers introduce real danger.
- Player must manage errors; this is a clear step up from the forgiving early levels.

6. Level Diagram (Textual)

BODY MAP – LEVEL 3

[Heart]

|

(warning)

|

v

[Liver] --performance--> [Immune Battlefield]

|

|

(toxins processed)

v

5. Level 4 – Hard

1. Narrative Setup

- Later the same week: the teenager has experienced several days of stress, poor sleep, and inconsistent eating.
- A serious infection plus high stress create a complex challenge.
- Mood: tense; music is faster, lighting is slightly darker in many areas.

2. Player Objectives

- Maintain a healthy state across multiple systems at once:
 - Advanced Brain (stress management).
 - Lungs (breathing rhythm).
 - Full Immune System (three enemy types).
- Prevent total collapse of Energy, Mood, and Health.

3. Detailed Content

3.1 Organs and Areas

- Advanced Brain – Control Room with multiple dials for Stress, Focus, and Motivation.
- Lungs – two large chambers with air flow currents and pulsing walls.
- Immune Command Center – central hub from where immune cells are dispatched.

3.2 New Enemy Types and Systems

Enemy types:

1) Virus:

- Small, very fast.
- Deals damage to Health if it touches the “Tissue Core” even once.

2) Bacterium:

- Medium speed, high health.
- Requires 2 immune tags instead of one.

3) Parasite:

- Slow, large.
- If not destroyed within 5 seconds after first tag, splits into two smaller parasites.

Brain – Stress Balance Mini-game:

- The player sees 3 sliders: Sleep, Work, Fun.
- Events appear as cards: “Exam Tomorrow”, “Gaming Night”, “Extra Homework”.
- Player must assign limited “Time Points” to Sleep, Work, Fun per turn.
- If Sleep too low for multiple turns, Stress rises sharply and Mood drops.

Lungs – Breathing Rhythm:

- A flow line shows inhaling and exhaling cycles.
- The player must press a key when small indicators align with ideal inhale/exhale zones.
- Missing too many beats makes Immune cells weaker (slower movement).

Immune Command Center – Battle:

- Player alternates between commanding squads (top-down view) and controlling a single immune cell in critical moments.
- Several “Tissue Nodes” must be defended from enemies.
- When a node falls, Health Bar takes a large hit.

3.3 Integrated Resource Effects

- High Stress in Brain increases the spawn rate of viruses.
- Poor performance in Lungs reduces movement speed of all immune cells by ~20%.
- Good resource balance (Sleep + moderate Work + some Fun) slowly regenerates Mood, making the game visually brighter.

3.4 Example Gameplay Flow

1) Player starts in Advanced Brain:

- Plays 3–4 turns of Stress Balance; early decisions affect starting enemy intensity.

2) Player moves to Lungs:

- Plays 2 breathing cycles of 30–40 seconds each.

3) Player enters Immune Command Center:

- Defends 3 Tissue Nodes for ~2–3 minutes of combined waves.

4) Events continue to cross-influence each other; e.g., losing a node increases Brain Stress again.

4. Win / Lose Conditions

Win:

- All Tissue Nodes survive the final wave.
- Health Bar stays above zero.

- At least neutral Mood at the end (not fully depleted).

Lose:

- Health Bar drops to zero.
- All Tissue Nodes are destroyed.
- Stress bar in Brain stays maxed out for too long (causes “burnout” event → auto-fail).

5. Difficulty Rationale

- Multiple systems interacting at once – player must think strategically, not just react.
- Fast enemies, split enemies, and slower immune cells create complex micro-management.
- Represents the cumulative effect of long-term bad habits and shows how hard it is to recover.

6. Level Diagram (Textual)

BODY MAP – LEVEL 4

[Advanced Brain]

|

(Stress decisions)

|

v

[Lungs]

|

(Breathing performance affects

Immune speed globally)

|

v

[Immune Command Center]

/ | \

[Tissue Node A] [Tissue Node B] [Tissue Node C]

|

Exit (only if at least 1 node survives)